Feature Scaling and Encoding

Feature Scaling

It is defined as a method to limit the range of the variables so that they can be compared on same grounds.
Feature scaling is a technique used in machine learning to preprocess data. It involves rescaling the features of your data to a common range.

<u>Data preprocessing</u>

- Anomalies.
  1. Outliers.
  2. Structure.
  3. + , , .
  4. Type
- Missing Values.
- Feature Scaling.

The example of the feature scaling is as follows:
Here's how feature scaling helps:

Feature 1: Years of Experience (1-10 years)
Feature 2: Online Coding Test Score (1-1000 points)
Without scaling, the score might dominate the decision because it has a much larger range.

Feature scaling puts them on a similar level playing field.
Scaled Years of Experience (0-1 range)
Scaled Online Coding Test Score (0-1 range)

Now, both features contribute equally, leading to a more balanced evaluation based on both experience and test performance.

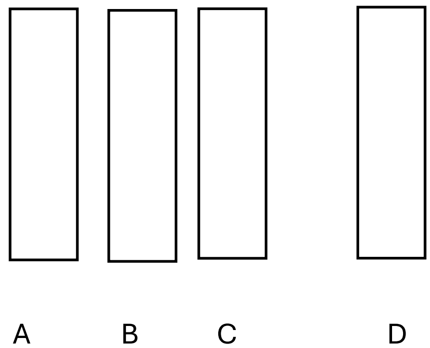
Why the data preprocessing is important?

Data preprocessing is crucial because it cleans and prepares raw data for analysis. Imagine a chef prepping ingredients - dirty or inconsistent data is like unwashed vegetables. Preprocessing ensures all features are on the same level playing field, leading to more accurate and reliable results from your analysis, just like following a recipe carefully yields a delicious meal.

Features

Let we have the dataset named as df:

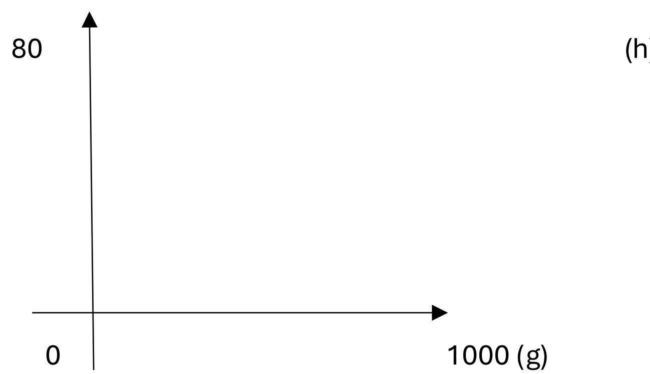
df =



Features **(Independent)** Labels **(Dependent)**

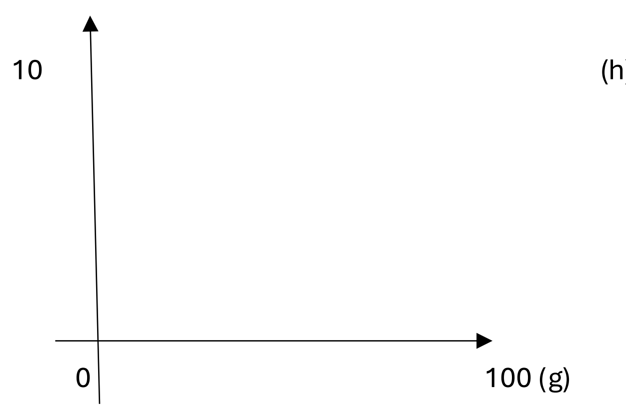
X = features = 1,2 or more than 20.
Y = labels = dependent.
For example, let we take the data of over own,

Food	Weight
600	70
700	71
800	72
900	73



Now after we have scaled all we have so:

Food	Weight
60	7.0
70	7.1
80	7.2
90	7.3



Methods to do the feature scaling is as follows:

1. Min-max scaler
2. Standard scaler
3. Reverse scaling
4. Logarithmic scaling

Library use for the scaling is the scikit-learn. We do the feature scaling only for the numeric variables.

1. **Min max scaling**

$$X` = \frac{x - \min(x)}{\operatorname{Max}(x) - \min(x)}$$

X = original value
X ` = scaled value.
In the end all the values turned out to be in a range in between 0 to 1.

2. **Standard scaling or z score normalization**

$$X` = \frac{x - \operatorname{mean}(x)}{\operatorname{Sd}(x)}$$

It ranges from -3 to +3.
If – 1 to +1 than values cover is 68%.
If -2 to +2 then values cover is 95%.
If -3 to +3 then values cover is 99.7%.

3. **Robust scaling**

$$X` = \frac{x - \operatorname{median}(x)}{\operatorname{IQR}(x)}$$

4. **Logarithmic scaling**

$$X` = \log(x)$$

Feature encoding

Feature encoding is the process of transforming categorical data into a numerical format that machine learning models can understand. This is important because many machine learning algorithms can only work with numerical features.

The important thing to know about is that it can only be done on the categorical variables.
In the titanic dataset in the seaborn library, we have variables that are:

Sex	Class
Male	First
Female	Second
	Third

Now these are the values type in the two variables of the titanic dataset, now to encode them we have so, that is:

Sex	Class
1	0
0	1
	2

The methods we must do the feature encoding is as follows:

1. One hot encoding.
2. Label encoding.
3. Rodinal encoding.

<u>One hot encoding</u>

Sex
M
F

Sex
T
F

<u>Label encoding</u>

Sex
M
F

Sex
1
0

<u>Ordinal encoding</u>

Convert it into meaningful order.

Class
First

Second
Third

Class
0
1
2

Why is the feature encoding important?

Feature encoding is crucial for machine learning because:

1. **Machines understand numbers:** Algorithms work best with numerical data, and encoding translates categories into numbers.
2. **Unlocks hidden patterns:** Numbers allow models to identify relationships between categories and other features.
3. **Improves model performance:** Encoding helps models learn from your data more effectively, leading to better predictions.